Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Methanamide (CHONH2)
Basis set : cc-pVDZ
Charge : 0
Multiplicity : 1
Total e- : 24
n-body order : 2

GROUND STATE ENERGIES

E0 (Ha) : -168.96056339
E1 (Ha) : -168.72195263
Gap (Ha) : 0.23861076
HF (Ha) : -168.94623212

CONVEX-ROOF METHODS

Methods run : gcg
Restarts : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,5)

Frozen core orbs : 9
Frozen e- : 18
Active e- : 6
Active orbs : 5
Spin-orbitals : 10
Hilbert dim : 100

SPIN PROPERTIES

<S2> : 0.000000
S : 0.0000
<Sz> : 0.000000
Multiplicity : 1
Spin state : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.988948
<D> / site : 0.597790

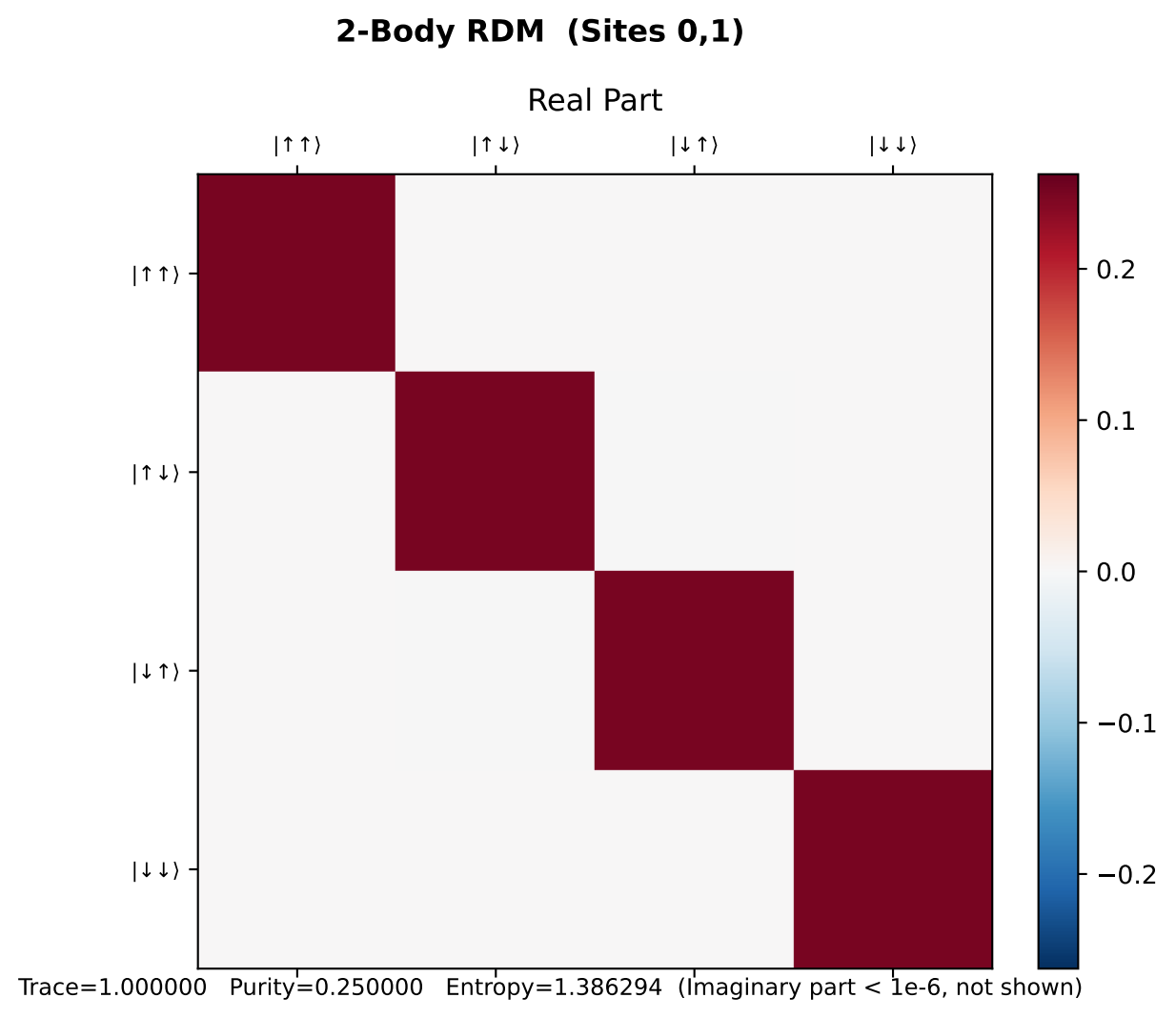
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Body order : 2
Clusters : 10

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Summary — 2-Body RDMs (Methanamide, CAS(6,5))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1	0.250000	—	—	—	—	—	MIXED
0,2	0.250000	—	—	—	—	—	MIXED
0,3	0.250170	—	—	—	—	—	MIXED
0,4	0.250590	—	—	—	—	—	MIXED
1,2	0.250000	—	—	—	—	—	MIXED
1,3	0.289364	—	—	—	—	—	MIXED
1,4	0.250074	—	—	—	—	—	MIXED
2,3	0.253576	—	—	—	—	—	MIXED
2,4	0.252556	—	—	—	—	—	MIXED
3,4	0.641191	—	—	—	—	—	WEAKLY MIXED


2-Body RDM (Sites 0,1)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,1)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000209
Von Neumann entropy	: 1.3862943193
Eigenvalue min	: 2.499583e-01
Eigenvalue max	: 2.501252e-01
Imaginary part	: < 1e-6 (not shown)

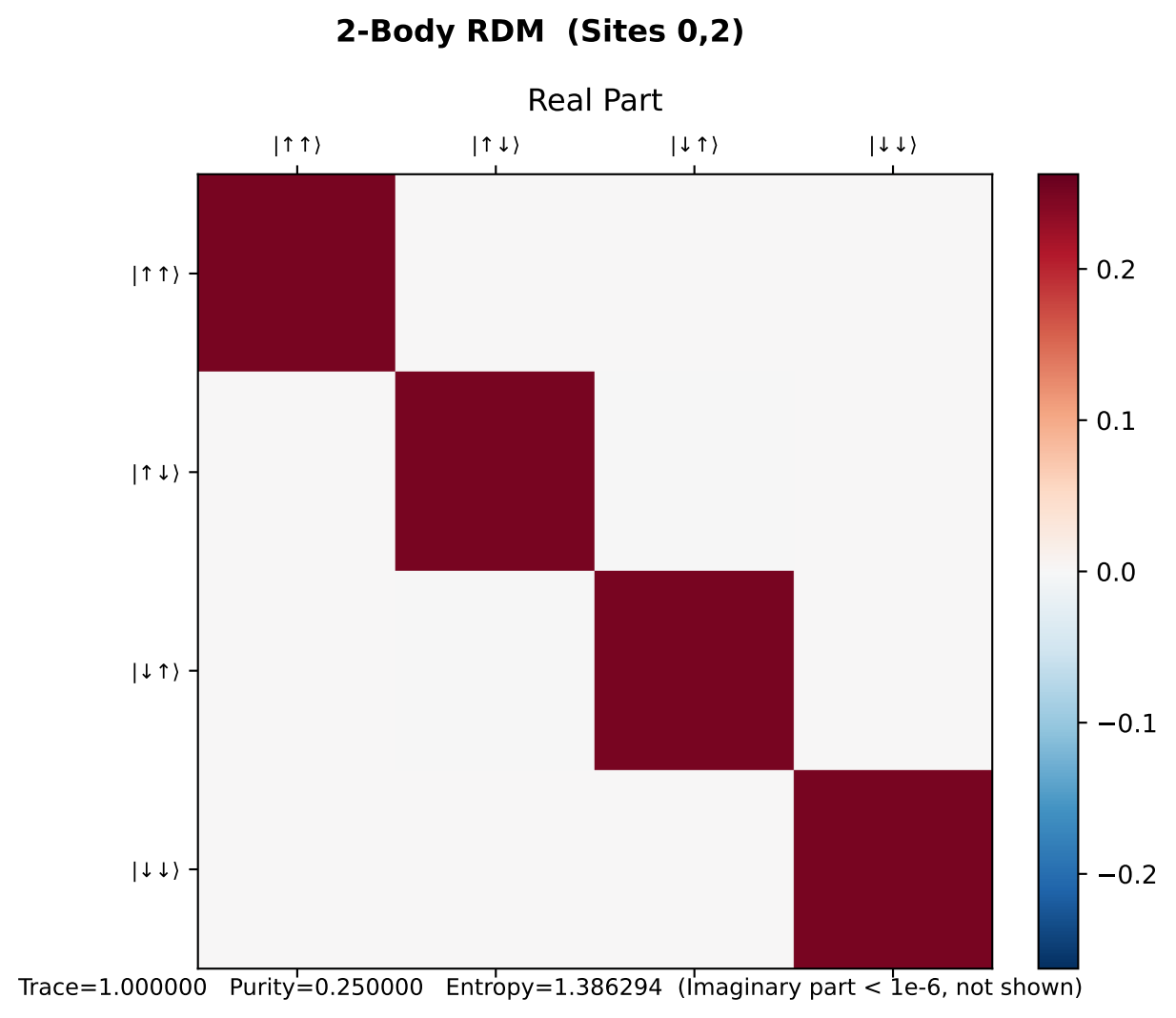
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000002

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,2)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500002246
Von Neumann entropy	: 1.3862939122
Eigenvalue min	: 2.498632e-01
Eigenvalue max	: 2.504104e-01
Imaginary part	: < 1e-6 (not shown)

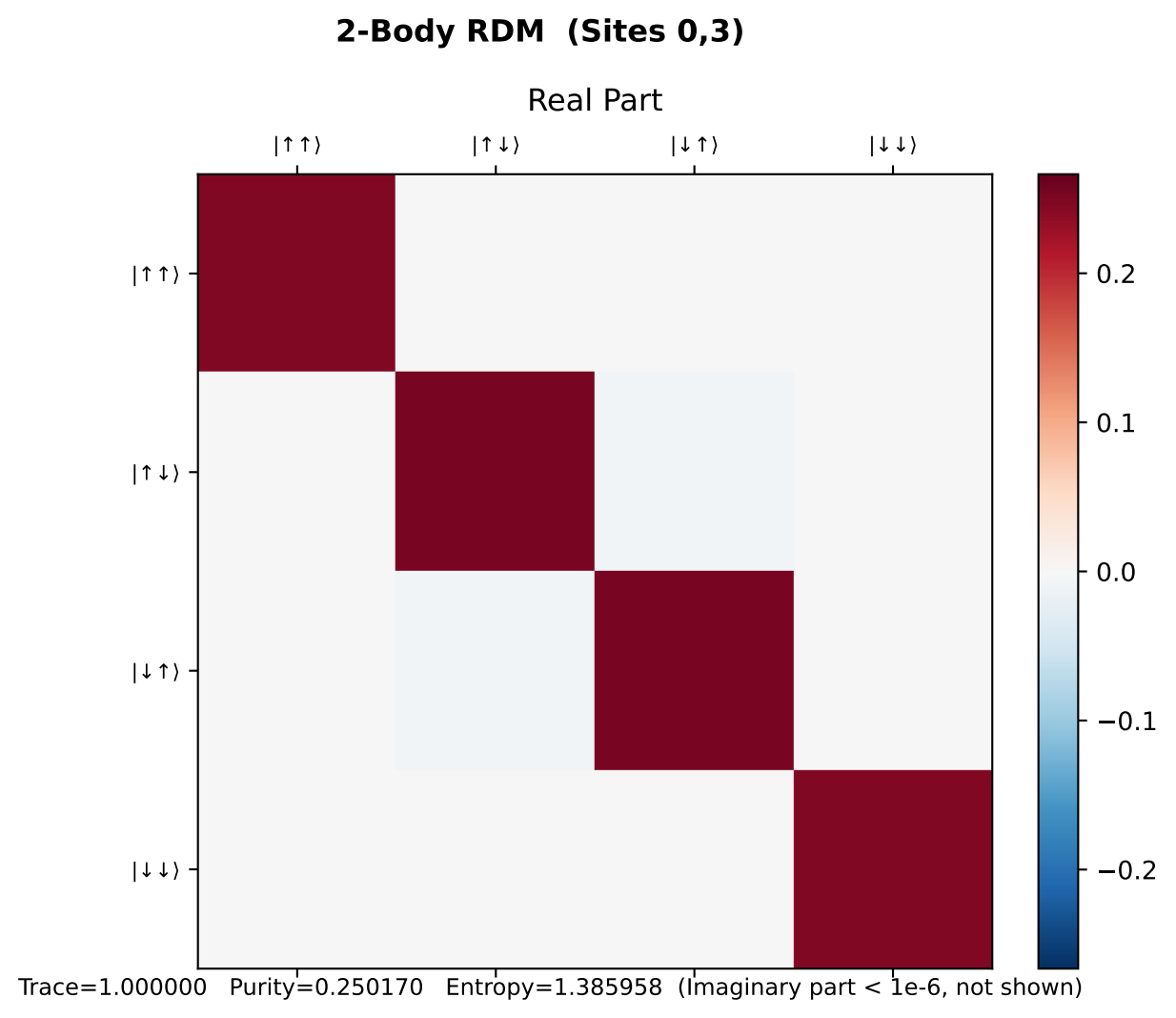
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000022

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250170 Entropy=1.385958 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2501700193
Von Neumann entropy	: 1.3859576480
Eigenvalue min	: 2.462359e-01
Eigenvalue max	: 2.612922e-01
Imaginary part	: < 1e-6 (not shown)

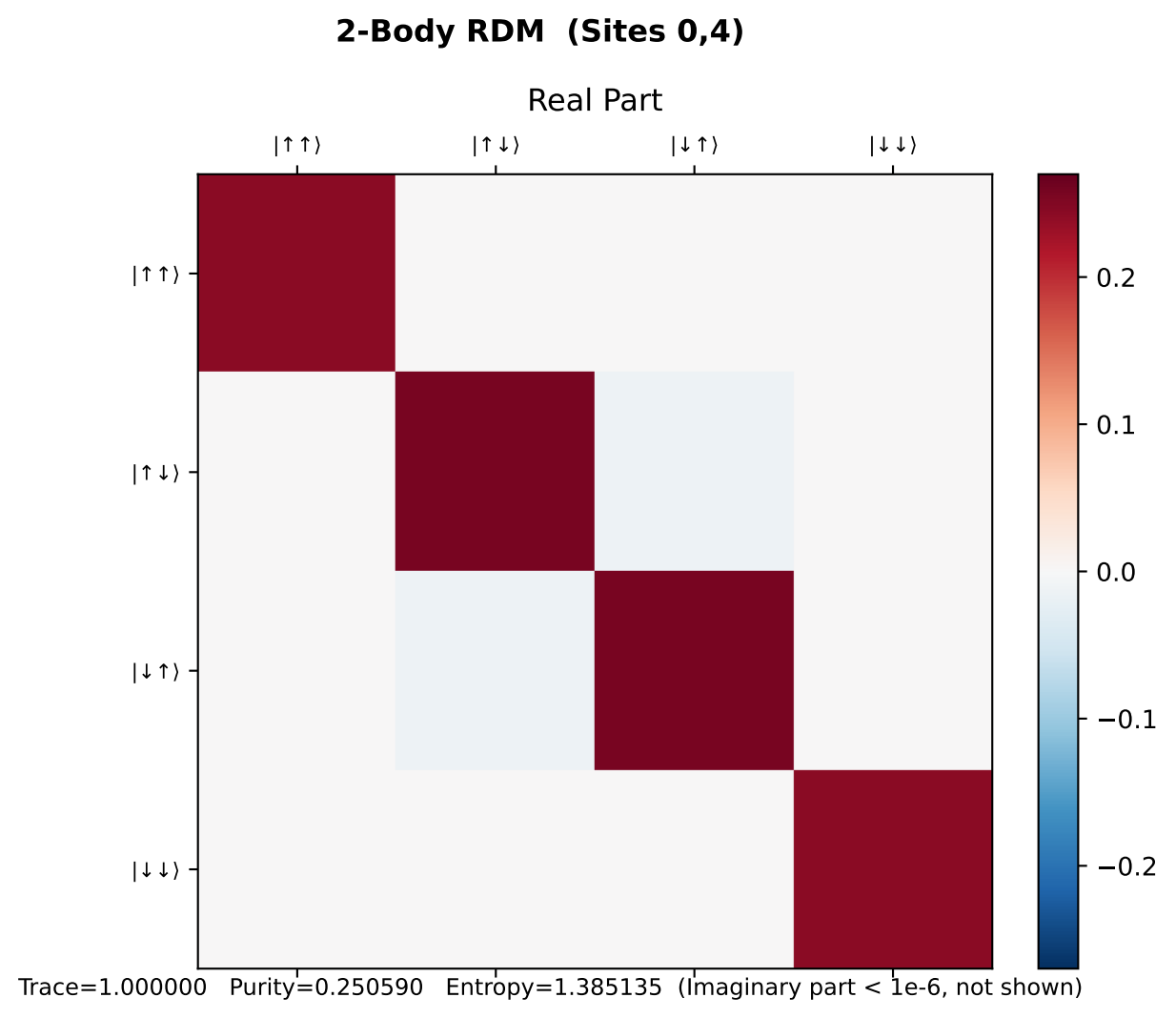
GME Analysis
==========

gamma = Tr(rho^2) = 0.25017002

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250590 Entropy=1.385135 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2505901026
Von Neumann entropy	: 1.3851351916
Eigenvalue min	: 2.429875e-01
Eigenvalue max	: 2.710375e-01
Imaginary part	: < 1e-6 (not shown)

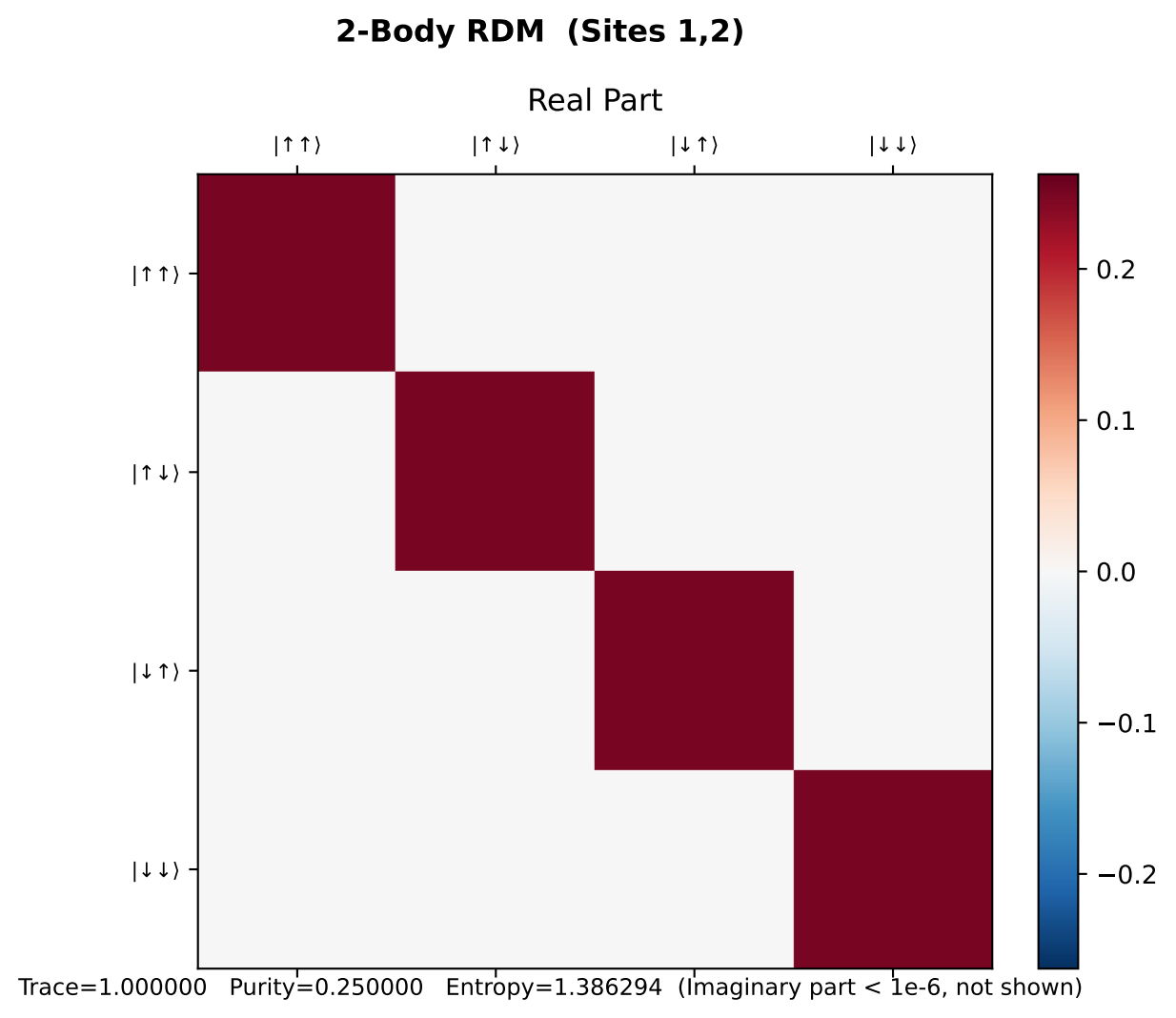
GME Analysis
==========

gamma = Tr(rho^2) = 0.25059010

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,2)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500001217
Von Neumann entropy	: 1.3862941178
Eigenvalue min	: 2.498993e-01
Eigenvalue max	: 2.503021e-01
Imaginary part	: < 1e-6 (not shown)

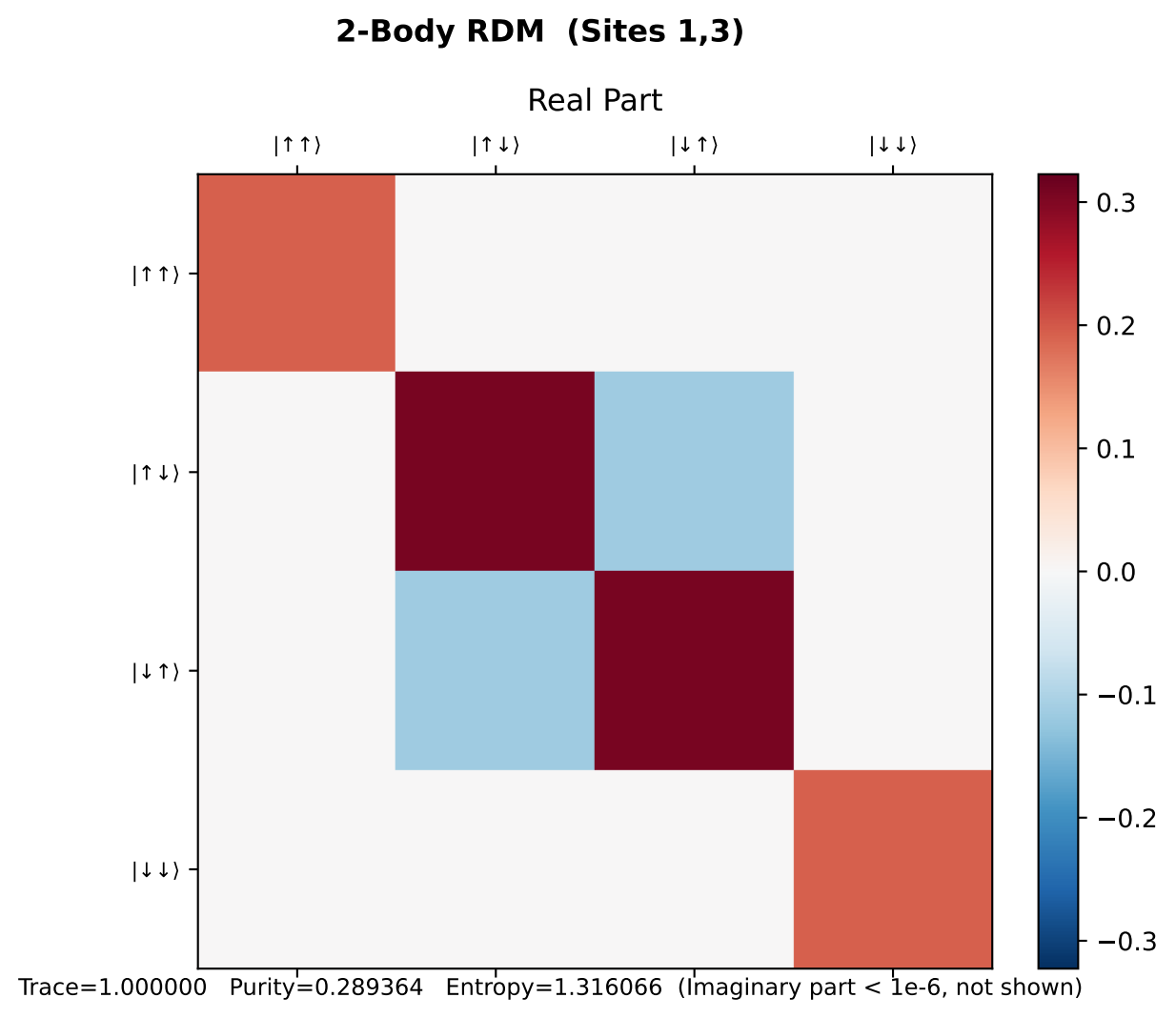
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000012

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=0.289364 Entropy=1.316066 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2893643176
Von Neumann entropy	: 1.3160657385
Eigenvalue min	: 1.927256e-01
Eigenvalue max	: 4.218233e-01
Imaginary part	: < 1e-6 (not shown)

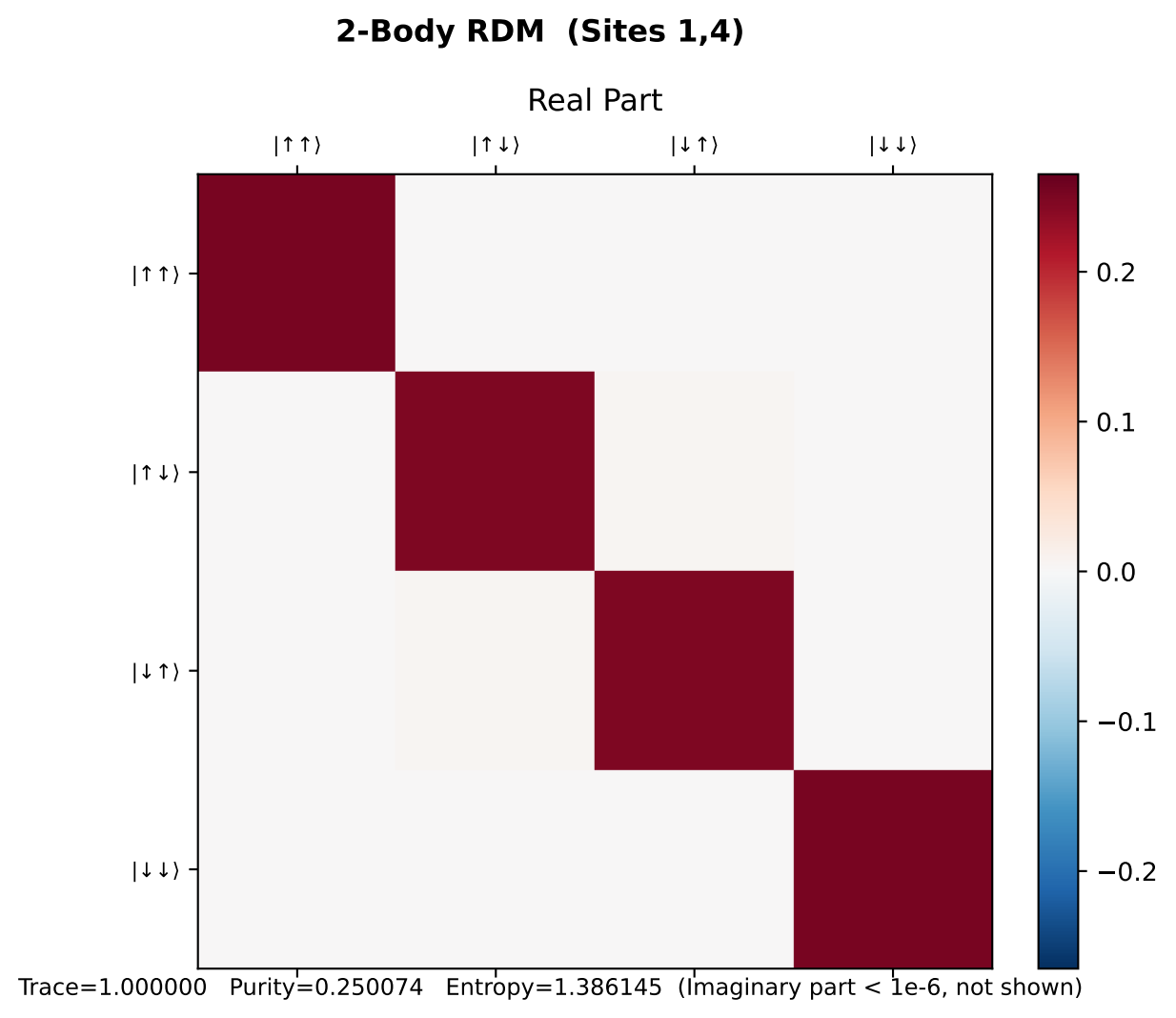
GME Analysis
==========

gamma = Tr(rho^2) = 0.28936432

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250074 Entropy=1.386145 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500741106
Von Neumann entropy	: 1.3861451403
Eigenvalue min	: 2.425446e-01
Eigenvalue max	: 2.524851e-01
Imaginary part	: < 1e-6 (not shown)

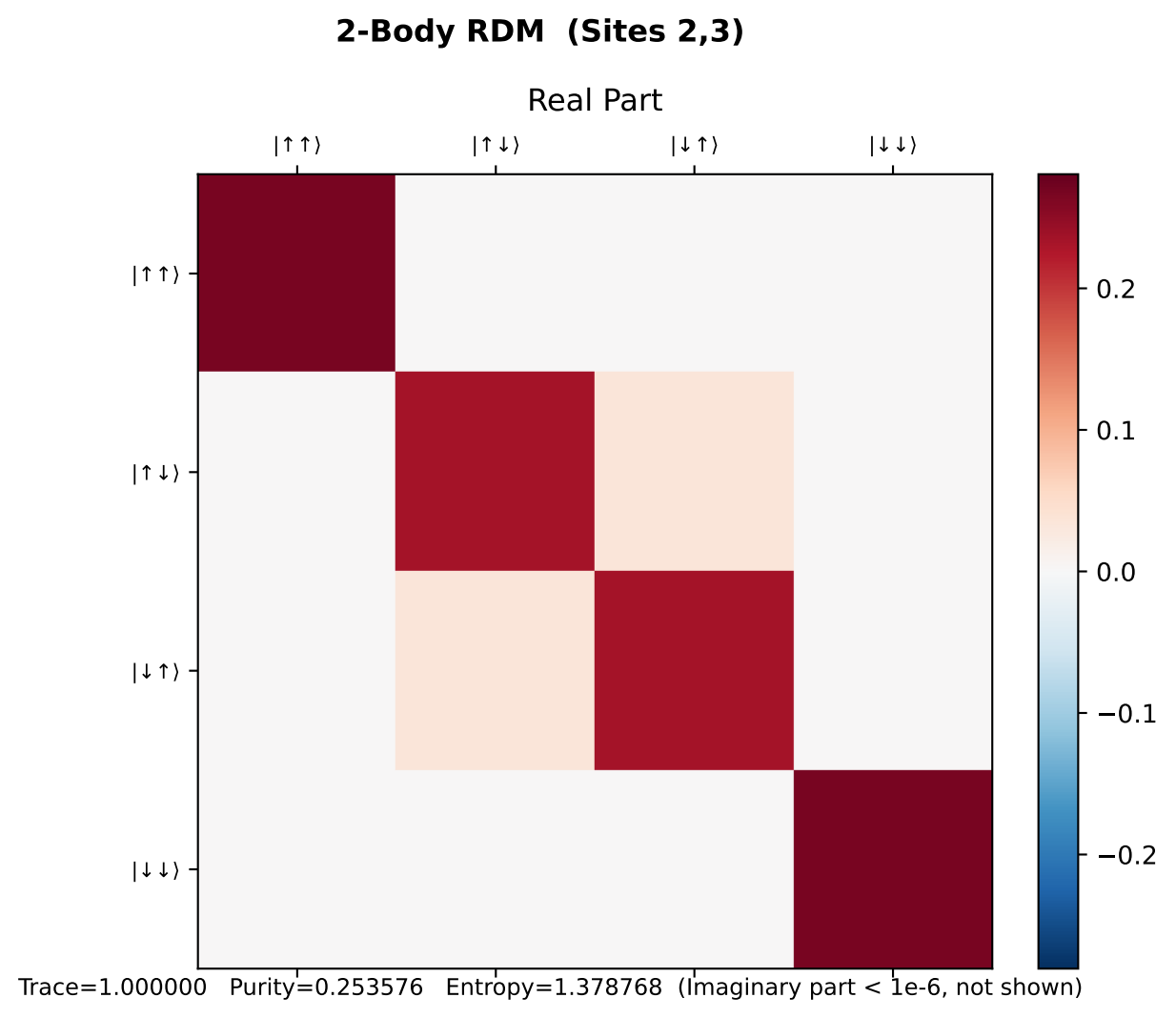
GME Analysis
==========

gamma = Tr(rho^2) = 0.25007411

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 2,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.253576 Entropy=1.378768 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 2,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2535761541
Von Neumann entropy	: 1.3787675238
Eigenvalue min	: 1.982109e-01
Eigenvalue max	: 2.672630e-01
Imaginary part	: < 1e-6 (not shown)

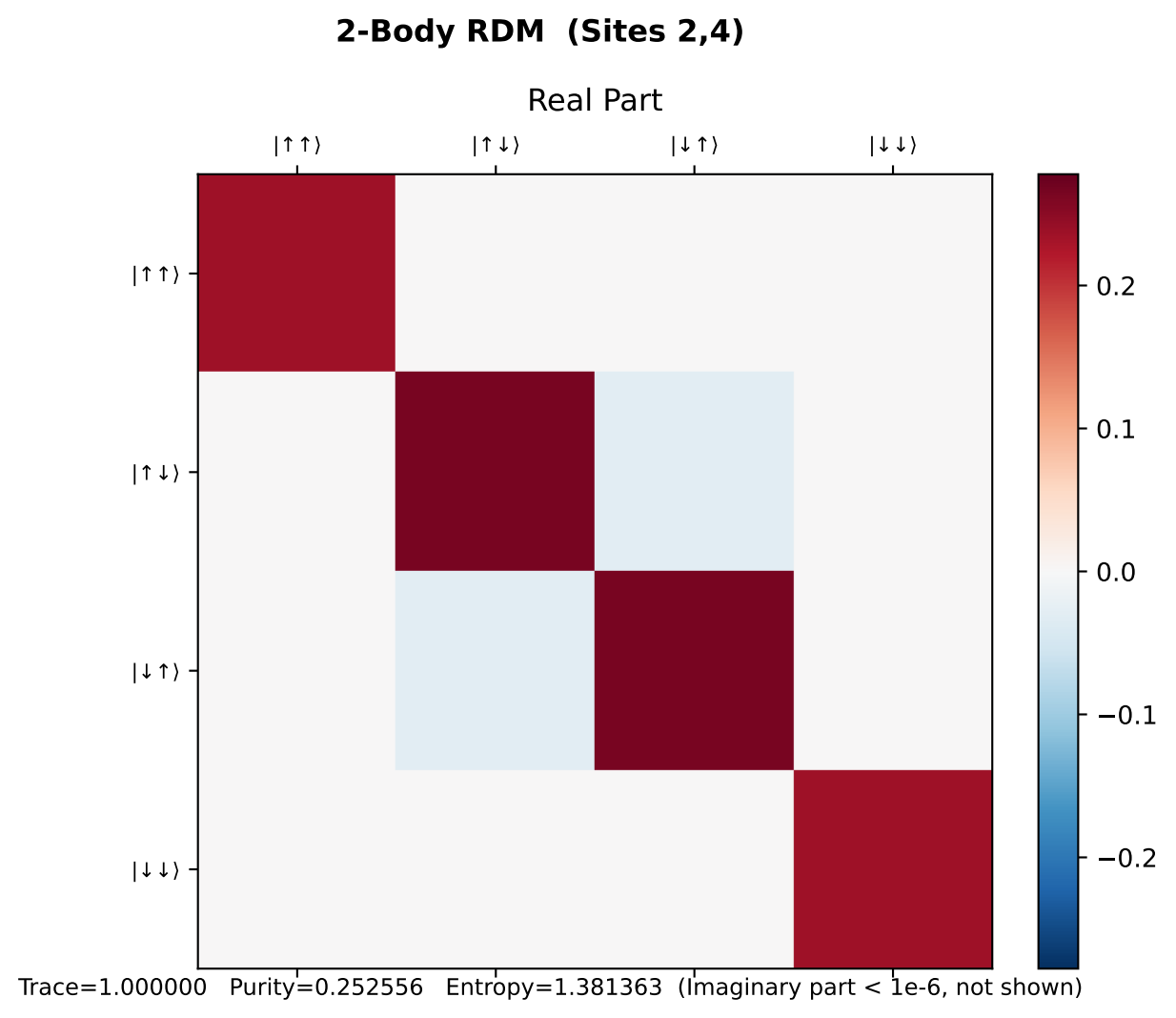
GME Analysis
==========

gamma = Tr(rho^2) = 0.25357615

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 2,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.252556 Entropy=1.381363 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 2,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2525559604
Von Neumann entropy	: 1.3813628826
Eigenvalue min	: 2.354056e-01
Eigenvalue max	: 2.937832e-01
Imaginary part	: < 1e-6 (not shown)

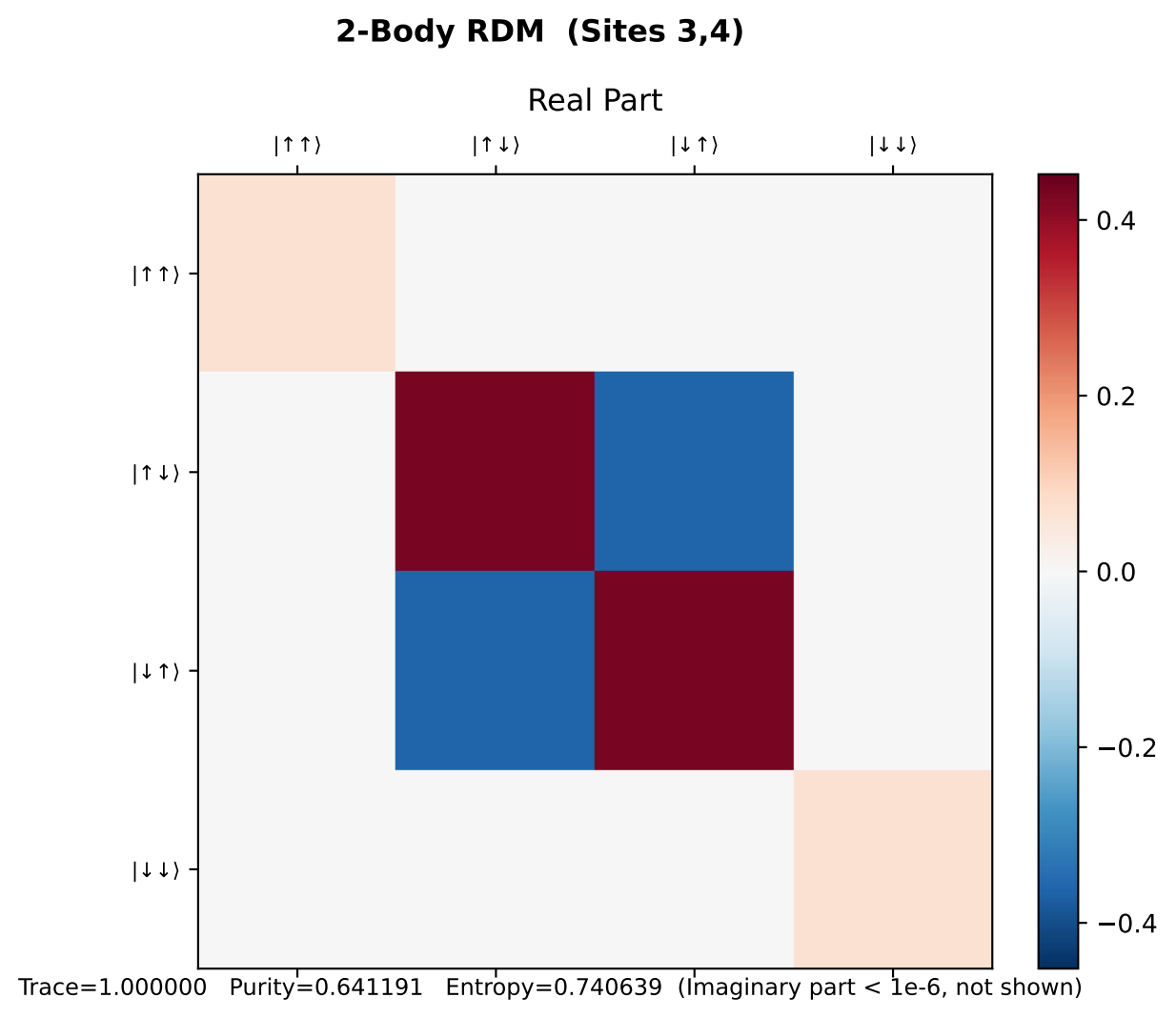
GME Analysis
==========

gamma = Tr(rho^2) = 0.25255596

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 3,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.641191 Entropy=0.740639 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.6411907646
Von Neumann entropy	: 0.7406394333
Eigenvalue min	: 6.944743e-02
Eigenvalue max	: 7.916577e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.64119076

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.